

NCEZID: Waterborne and Fungal Infections



Water is essential for life but can also lead to illness when it is contaminated by disease-causing organisms. Examples of **waterborne disease** include:

- Cholera, a severe diarrheal disease, often caused by drinking *Vibrio cholerae*-contaminated water.
- An infection called microbial keratitis, which can occur when germs invade the eye. These germs can be associated with improper use and care of contact lenses.
- Cryptosporidiosis, giardiasis, and other gastrointestinal diseases caused by swallowing contaminated water.
- An infection caused by the rare but often fatal brain-eating ameba (*Naegleria fowleri*) that can occur when people go swimming or diving in warm freshwater places like lakes and rivers, and contaminated water enters through the nose.

Related Resources

[Stay Healthy Features: Stay Safe Drinking—or Playing—Water](#)

[Waterborne Infections on Medscape](#)

[Waterborne Infections Videos](#)

[Water Safety Graphics](#)

[Waterborne Infections Blogs](#)

[Fungal Infections Graphics](#)

[Fungal Infections Blogs](#)

Fungal diseases are a public health problem because of:

- **Increasing number of people with weakened immune systems.** Opportunistic infections such as cryptococcal meningitis typically affect people living with HIV/AIDS and others with weakened immune systems. Other infections like aspergillosis most commonly affect people with cancer and those who have had organ transplants.
- **Changes in healthcare practices.** Hospital-associated infections such as candidemia are a major concern in the United States. Drug-resistant fungi are emerging in healthcare settings.
- **Changes in the environment.** Fungi that live in soil or other parts of the natural environment cause diseases such as coccidioidomycosis (Valley fever) and histoplasmosis. Weather and climate changes may be affecting these types of fungi.

What we're doing

NCEZID has world-class scientists who are experts in the fields of preventing and controlling waterborne and fungal diseases. Recent accomplishments include:

- Tested more than 111 clinical samples in the US for **free-living amebas**, including *Naegleria fowleri*, the "brain-eating" ameba.

- Implemented **CryptoNet**, a system that uses molecular fingerprints for tracking a parasite. *Cryptosporidium* is a major cause of US waterborne disease outbreaks.
- Released the 3rd edition of the **Model Aquatic Health Code** to help state and local health departments prevent drowning, injuries, and the spread of recreational water illnesses at public swimming pools and spas.
- Provided epidemiological field support to ministries of health in multiple countries to prevent and **control outbreaks of cholera and typhoid**.
- Applied new **laboratory tools to detect valley fever** (coccidioidomycosis) in the environment and used whole genome sequencing to understand the emergence of this fungus in new geographic areas.

Investigating **new cases of *Candida auris* infections**, an emerging drug-resistant fungus that can spread through healthcare settings and cause serious and sometimes deadly bloodstream infections that have limited treatment options.

Stories on Waterborne & Fungal Infections

[Partnership in Action: Healthier and Safer in the Water](#)

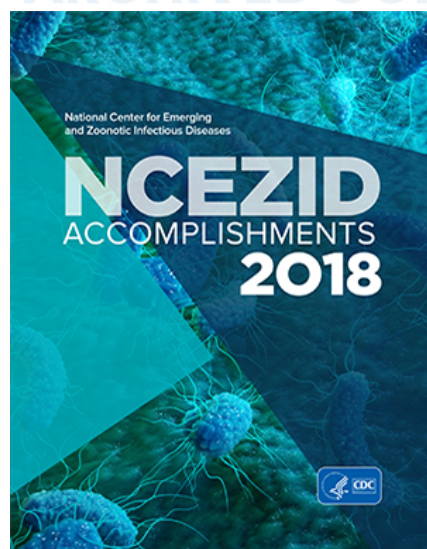
[Partnership in Action: NCEZID Partners with CDC Foundation and Pfizer on Cryptococcal Screening in South Africa](#)

[Global Story: CDC Responds to a Dangerous Fungus Here and Abroad](#)

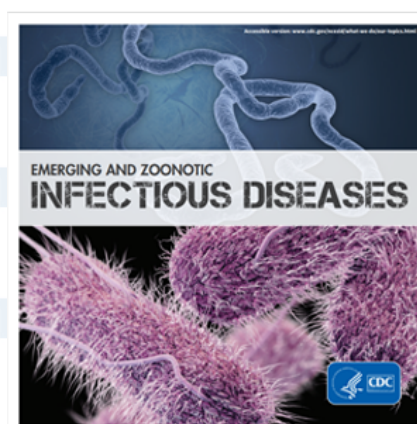
[Global Story: NCEZID Applied Epidemiology Training Helps Cholera Outbreak Response in Kenya](#)

Accomplishments & Innovations

Find more information on accomplishments and innovations in Waterborne and Fungal Infections in these publications.



[NCEZID Accomplishments 2018](#)



[Emerging and Zoonotic Infectious Diseases](#)



[Innovations to Stop Emerging and Zoonotic Infections](#)

Last Reviewed: December 17, 2019

Source: Centers for Disease Control and Prevention , National Center for Emerging and Zoonotic Infectious Diseases (NCEZID)